

J-space Phase 4 — development synthesis

OLMo training trajectory and coordinate-frame validity

State through OLMo-3 32B base and 3.0 Think · 2026-07-31

Scope. This is a living **development only** report over known Phase 3 banks. It localizes trajectory patterns and audits lens-frame dependence; it cannot establish a binary lineage claim. No Phase 4 confirmatory or replication cell is licensed before PI sign-off and a tagged preregistration freeze.

1 Compute and provenance boundary

Every model producer hard-failed unless CUDA was visible in the same process, executed a finite FP16 CUDA matrix multiply before model load, and recorded the device in its result envelope. Jobs ran on an NVIDIA RTX PRO 6000 Blackwell Server Edition; the final grid allocated about 81.1 GB VRAM. Model loading, generation, interventions, and scoring never used CPU. CPU was limited to tests, hashes, statistics, figures, and document compilation.

Phase 3 enters through the immutable `p4-import-phase3-release-v1`. Native evidence is event-sourced; registered artifacts are never overwritten.

2 Prospective capability gate

OLMo-3 32B Think produced 972 G5 rows, 324 facts, and 72 families under prospective prefix-disjoint accepted-alias scoring.

view	overall	Bank F	Bank S	direct	composed	bridge supplied
capability	.6430	.4935	.8972	.6574	.4846	.7870

The intervention cohort was fixed before outcomes by requiring both direct and composed generation capability within a fact: 41/204 Bank-F facts and 85/120 Bank-S facts, or 126 facts / 252 items.

Evidence: `p4-g5-bank-olmo3-think-dev-v2`

The first attempt stopped after accent folding made `Río` and `Rio` ambiguous. The repaired selector prefers exact spelling before a unique normalized fallback; the partial attempt remains preserved.

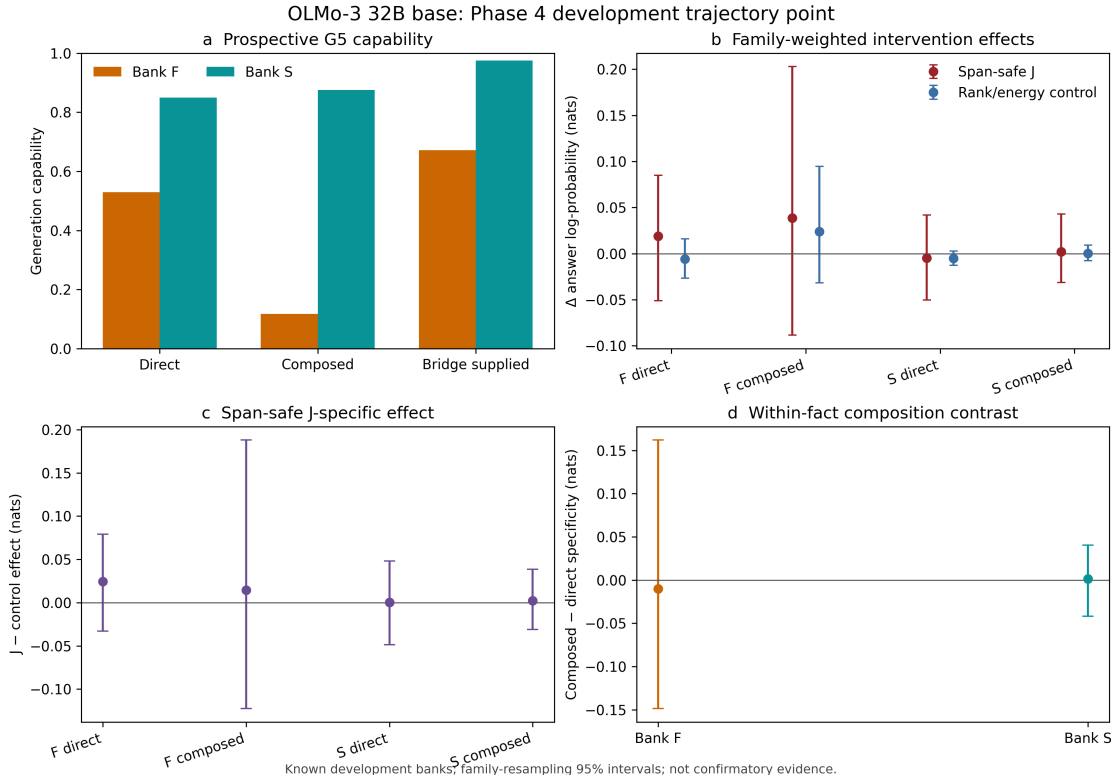
3 OLMo-3 32B base point

The exact pinned base revision `c2b61dae89a1ad10e4ad5653d0e46b590902607b` produced 972 G5 rows. Overall capability was .6101 (Bank F .4395; Bank S .9000). Requiring both direct and composed capability fixed 23 Bank-F and 88 Bank-S facts, or 111 facts / 222 items.

Evidence: `p4-g5-bank-olmo3-base-dev-v1`

The seven-condition grid had maximum baseline replay drift 6.71×10^{-7} nats, zero span-safe selected/protected overlap, 100% matched-rank agreement, and maximum energy relative error 0.000260.

Evidence: `p4-lineage-grid-olmo3-base-dev-v1`



cell	J-specific estimate (nats)	family-bootstrap 95% interval
Bank F direct	+0.0245	$[-0.0330, +0.0792]$
Bank F composed	+0.0145	$[-0.1226, +0.1881]$
Bank S direct	+0.0005	$[-0.0487, +0.0481]$
Bank S composed	+0.0021	$[-0.0309, +0.0385]$

All four primary effects are near zero. Bank-S composed-minus-direct is $+0.0016 [-0.0417, +0.0403]$. The experiment is nevertheless sensitive: overall label-protected J-specific is $+0.1231 [+0.0667, +0.1802]$, while mechanics-random and logit-protected effects are -0.1048 and -0.2023 , with intervals below zero.

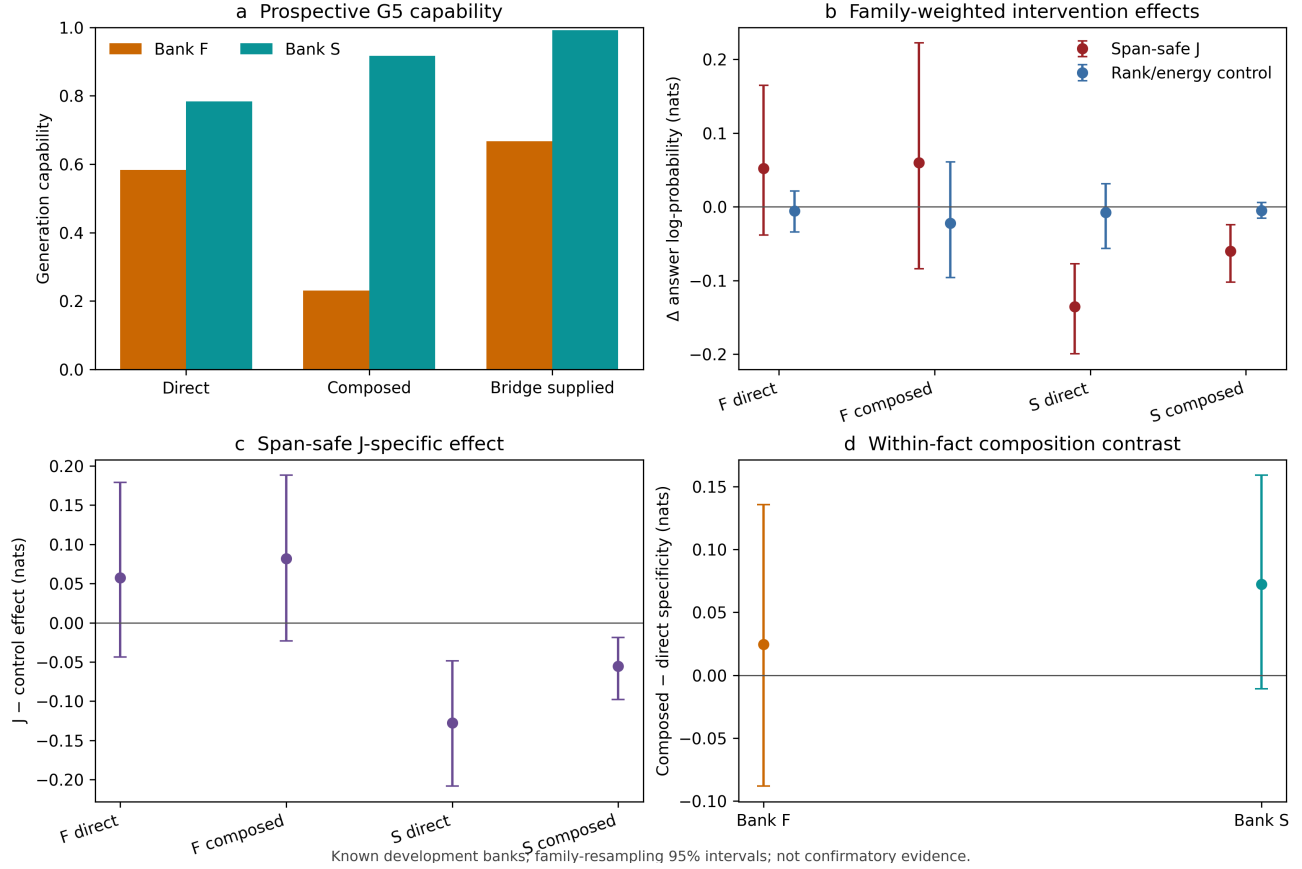
Evidence: [p4-lineage-analysis-olmo3-base-dev-v1](#)

4 Think own-lens development point

The seven-condition grid includes baseline, span-safe J, exact rank/energy control, label-protected J, protected-energy control, mechanics random, and logit-space label protection. Baseline replay drift is at most 7.15×10^{-7} nats. Rank agreement is 100%, energy relative error is at most 0.000230, and span-safe selected/protected overlap is zero.

Evidence: [p4-lineage-grid-olmo3-think-dev-v1](#)

OLMo-3 32B Think: Phase 4 development trajectory point



cell	J-specific estimate (nats)	family-bootstrap 95% interval
Bank F direct	+0.0574	[−0.0442, +0.1785]
Bank F composed	+0.0818	[−0.0223, +0.1878]
Bank S direct	−0.1277	[−0.2072, −0.0494]
Bank S composed	−0.0553	[−0.0982, −0.0187]

Bank-S composed-minus-direct specificity is +0.0724 [−0.0109, +0.1584]. It is compatible with the unusual 3.1 Think tendency beginning by 3.0, but is not resolved and must await the controlled Bank-W redundancy/derivation study. Evidence: [p4-lineage-analysis-olmo3-think-dev-v1](#)

5 Common-base-lens repair audit

The common coordinate is the frozen OLMo base lens (92f32e...b696). V1 exposed rank-limited protected-energy sites that omitted one requested component without being marked clamped. V2 repaired the metadata but revealed that the evidence ID itself changed all matched/random seeds.

V3 freezes the scientific-seed namespace to v1. All seven aggregate outcome columns, every per-alias score, and every condition order match v1 exactly (maximum difference 0). The fix reclassifies 296 summaries and reduces the erroneous maximum unclamped energy error from 0.511283 to 0.000239 while retaining 100% rank agreement. V3 supersedes both withdrawn diagnostic attempts.

Evidence: [p4-lineage-grid-olmo3-think-common-base-lens-dev-v3](#)

6 Own/common lens comparison: repair pending

The v1 paired analysis is **withdrawn**. Although the two raw grids pair 252 items and have zero baseline drift, their scientific seed namespaces differ. Their matched-control random subspaces therefore differ across frames. For Bank F

composed, the control shift alone is +0.0616 nats with its interval above zero, so the previous frame-contrast intervals and correlations mix lens change with RNG change and must not be interpreted.

Evidence: `p4-lens-frame-analysis-olmo3-think-dev-v1(withdrawn)`

Both raw grids remain valid standalone development evidence:

cell	own-lens point estimate	common-base-lens point estimate
F direct	+0.0574	−0.0403
F composed	+0.0818	−0.0656
S direct	−0.1277	−0.0861
S composed	−0.0553	−0.0464

A common-lens v4 grid is configured with the own-grid namespace. The hardened analysis now refuses unequal namespaces. No inferential frame comparison is live until v4 and the replacement v2 analysis pass their acceptance audits.

7 Interpretation and next boundary

1. The base checkpoint is near zero in all four primary cells. Bank-S direct and composed specificity is negative by 3.0 Think and is negative in each standalone lens frame; formal frame stability awaits the seed-paired repair.
2. Because capability cohorts are checkpoint-specific, this localizes a base-to-Think trajectory change but is not a paired causal estimate.
3. Bank-F standalone point estimates differ by frame, but the prior paired inference is withdrawn because its controls used different RNG streams.
4. The positive Bank-S composition tendency remains uncertain.
5. None of these known-bank estimates licenses “lineage present” or “lineage absent.”

The pinned base checkpoint is complete and banked. First run the seed-paired 3.0 common-lens repair and replacement analysis. Next, run the exact 3.1 Think and 3.1 Instruct checkpoints through the Phase 4 prospective G5 gate and the same seven-condition GPU grid, explicitly sharing an RNG namespace within each own/common pair. Then synthesize the full base → 3.0 Think → 3.1 Think/Instruct trajectory. If the two coordinate views disagree, prioritize the fit-size/corpus study before stronger causal interpretation.